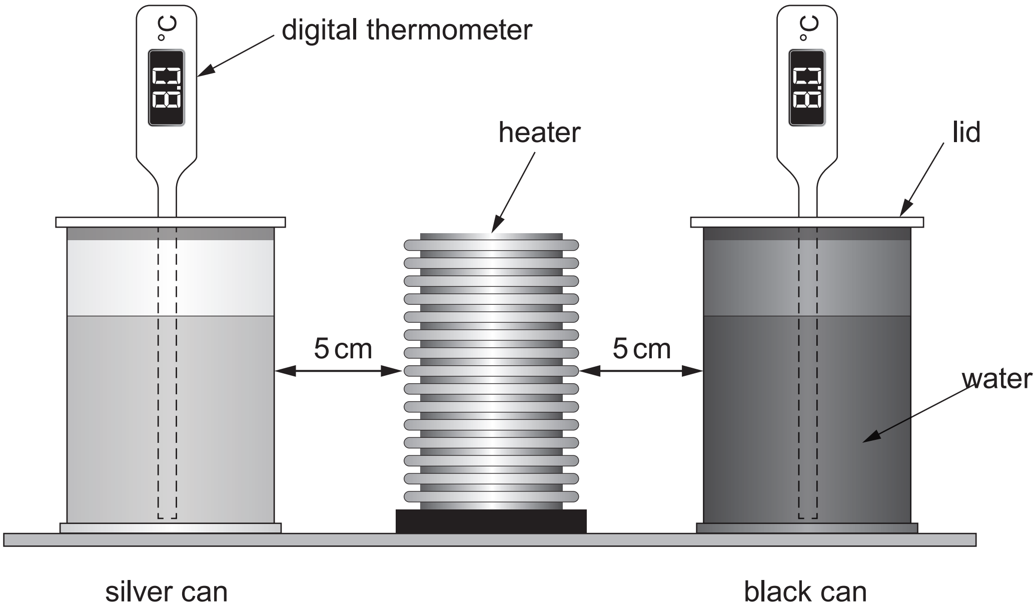
Examiner only

5. A pupil sets up an experiment to investigate the absorption of heat radiation.

The diagram shows the apparatus set up **before** the heater was turned on for 8 minutes.

Two identical aluminium cans were both filled with 200 cm^3 of water and a digital thermometer was inserted through a small hole in the lid.

One can was painted black and the other painted silver.



(a) In the experiment several variables were controlled in order to make a valid conclusion.

Use the information above to describe how this was achieved.

In your answer include
- **three** variables that were controlled
- an explanation of why each was controlled. [6 QER]

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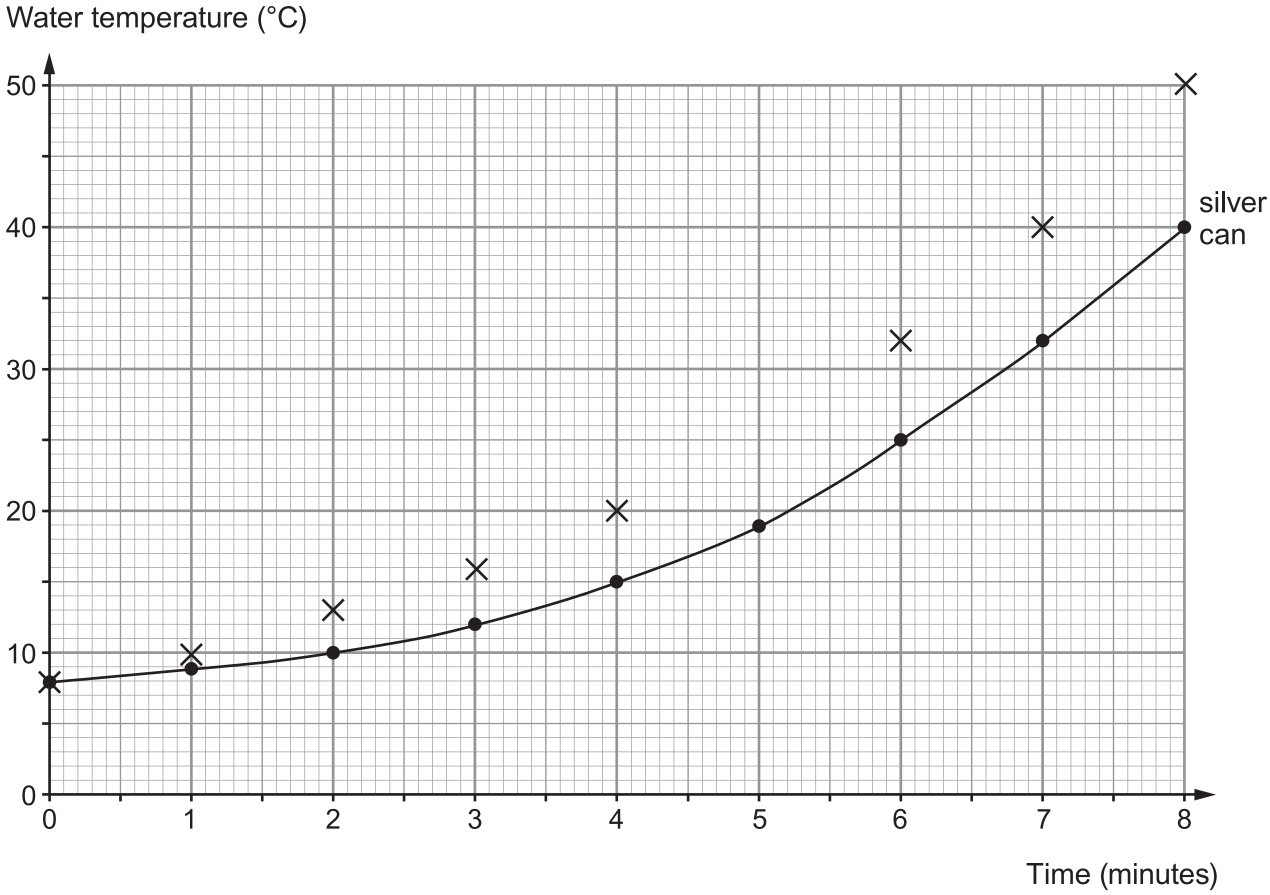
(b) Each minute, the temperature of the water in each can was measured.

The data, for the silver can, have been plotted on the grid below and its best fit curve drawn.

Apart from one result, the data points for the black can were also plotted on the grid.



(i) At **5 minutes** the water temperature in the black can was **25 °C**.

Plot this missing data point on the grid above **and** add the line of best fit. [2]

(ii) Compare the temperature rise in the black can with the temperature rise in the silver can.

Explain your answer in terms of heat radiation. [2]

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